	AIR UNIVERSITY
	DEPARTMENT OF COMPUTER ENGINEERING
	EXPERIMENT NO: 4

Lab Tittle: Basic Router Configuration

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Objectives: Properly cable and configure a network according to the topology diagram. Reset routers to their default state and erase existing configurations. Perform initial configurations on routers, including hostname and password settings. Configure IP addresses on router interfaces and host PCs. Verify router configurations and test network connectivity using ping and show commands. Analyze the network setup and troubleshoot any issues encountered.

LAB ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Ability to Conduct Experiment					
Ability to assimilate the results					
Effective use of lab equipment and follows the lab safety rules					

Total Marks:_____ Obtained Mark: _____

LAB REPORT ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Data presentation					
Experimental results					
Conclusion					

Total Marks: _____ Obtained Marks: _____



DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING

Basic Router Configuration

Lab Report: 4

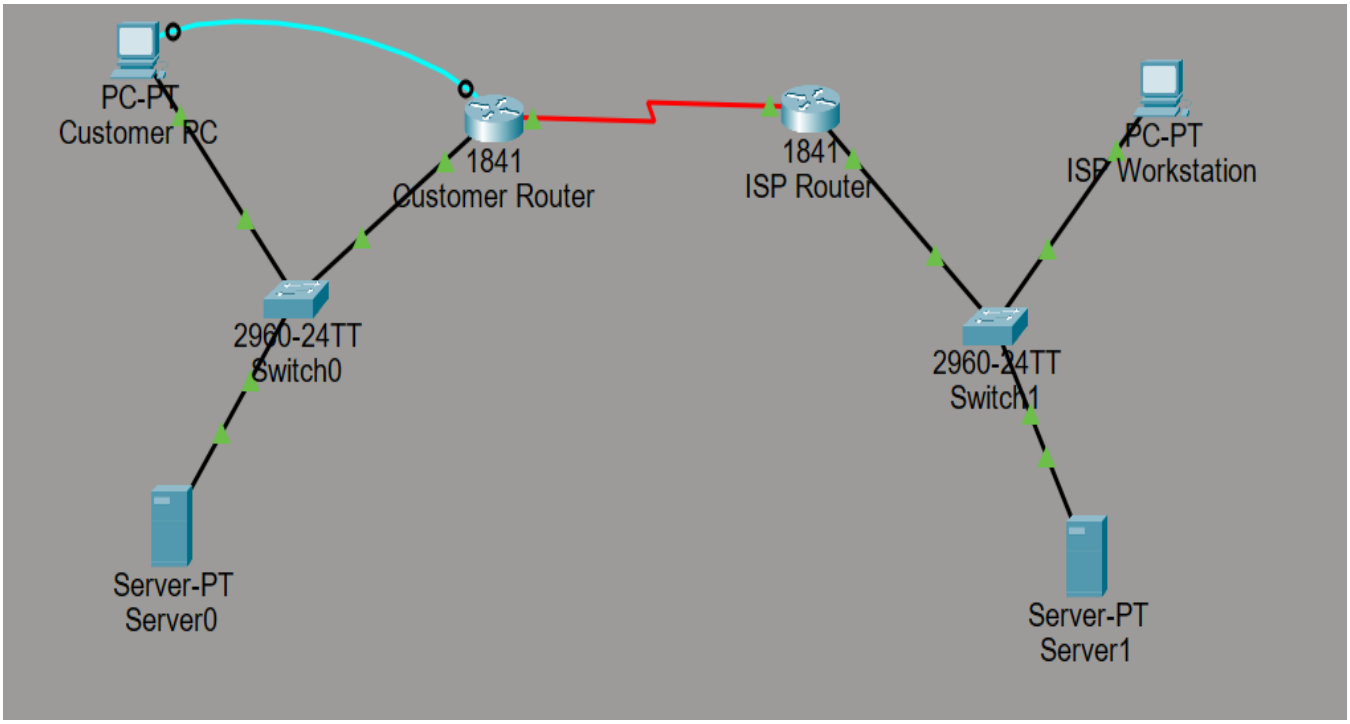
Tittle: Basic Router Configuration

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Submitted To: CE- Engr. Sir Basharat Ahmad

Date: 6 Oct, 2026



Customer Router

```
Device Name: Customer Router
Device Model: 1841
Hostname: User-Router

Port              Link   VLAN   IP Address         IPv6 Address                              MAC Address
FastEthernet0/0   Up     --     192.168.1.1/24     <not set>                                 0090.2B39.C10
FastEthernet0/1   Down   --     <not set>          <not set>                                 0090.2B39.C10
Serial0/1/0       Up     --     192.168.2.1/24     <not set>                                 <not set>
Serial0/1/1       Down   --     <not set>          <not set>                                 <not set>
Vlan1             Down   1      <not set>          <not set>                                 0003.E4E4.E72

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Customer Router
```

ISP Router

```
Device Name: ISP Router
Device Model: 1841
Hostname: Router-ISP

Port              Link   VLAN   IP Address         IPv6 Address                              MAC Address
FastEthernet0/0   Up     --     192.168.3.1/24     <not set>                                 0003.E48D.5E01
FastEthernet0/1   Down   --     <not set>          <not set>                                 0003.E48D.5E02
Serial0/1/0       Up     --     192.168.2.2/24     <not set>                                 <not set>
Serial0/1/1       Down   --     <not set>          <not set>                                 <not set>
Vlan1             Down   1      <not set>          <not set>                                 0004.9A48.67C6

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > ISP Router
```

1. **Configure the router host name.**
2. **Configure passwords.**
3. **Configure banner messages.**

Customer-Router Configuration

Physical | Config | CLI | Attributes

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname User-Router
User-Router(config)#enable password waq123
User-Router(config)#enable secret waq321
User-Router(config)#no ip domain-lookup
User-Router(config)#banner motd &
Enter TEXT message.  End with the character '&'.
!!!Authorized Access only &

User-Router(config)#line console 0
User-Router(config-line)#password 1234
User-Router(config-line)#login
User-Router(config-line)#exit
User-Router(config)#interface FastEthernet0/0
User-Router(config-if)#ip address 192.168.1.1 255.255.255.0
User-Router(config-if)#no shutdown

User-Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

User-Router(config-if)#interface serial 0/0/0
%Invalid interface type and number
User-Router(config)#interface serial 0/1/0
User-Router(config-if)#ip address 192.168.2.1 255.255.255.0
User-Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/1/0, changed state to down
User-Router(config-if)#exit
User-Router(config)#ip route 192.168.2.0 255.255.255.0 192.168.2.2
User-Router(config)#end
User-Router#
%SYS-5-CONFIG_I: Configured from console by console

User-Router#write
Building configuration...
[OK]
User-Router#
```

Verify the Customer-router configuration

```
User-Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        192.168.1.1     YES manual up                    up
FastEthernet0/1        unassigned      YES unset  administratively down down
Serial0/1/0            192.168.2.1     YES manual up                    up
Serial0/1/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
User-Router#show running-config
Building configuration...

Current configuration : 913 bytes
!
version 12.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname User-Router
!
!
!
enable secret 5 $1$mERr$Gvkz9a6HXkI2nY8cSO/Rf.
enable password waq123
!
interface FastEthernet0/0
 ip address 192.168.3.1 255.255.255.0
 duplex auto
 speed auto
!
interface FastEthernet0/1
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Serial0/1/0
 ip address 192.168.2.2 255.255.255.0
 clock rate 64000
!
interface Serial0/1/1
 no ip address
 clock rate 2000000
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
ip classless
ip route 192.168.1.0 255.255.255.0 192.168.2.1
!
ip flow-export version 9
banner motd ^C
!!Authorized access only!! ^C
!
!
!
!
!
line con 0
 password 1234
!
line aux 0
!
line vty 0 4
 password 1234
 login
```

ISP Workstation-Router Configuration

Physical | Config | CLI | Attributes

```
Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname IS-Router
IS-Router(config)#enable password 1234
IS-Router(config)#enable secret 4321
IS-Router(config)#no ip domain-lookup
IS-Router(config)#banner motd &
Enter TEXT message.  End with the character '&'.
!!Only Authorized Access only !! &

IS-Router(config)#line console 0
IS-Router(config-line)#password 1234
IS-Router(config-line)#login
IS-Router(config-line)#exit
IS-Router(config)#interface FastEthernet0/0
IS-Router(config-if)# ip address 192.168.3.1 255.255.255.0
IS-Router(config-if)#no shutdown

IS-Router(config-if)#
%LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to down

IS-Router(config-if)#interface serial 0/1/0
IS-Router(config-if)#ip address 192.168.2.2 255.255.255.0
IS-Router(config-if)#clock rate 64000
IS-Router(config-if)#no shutdown

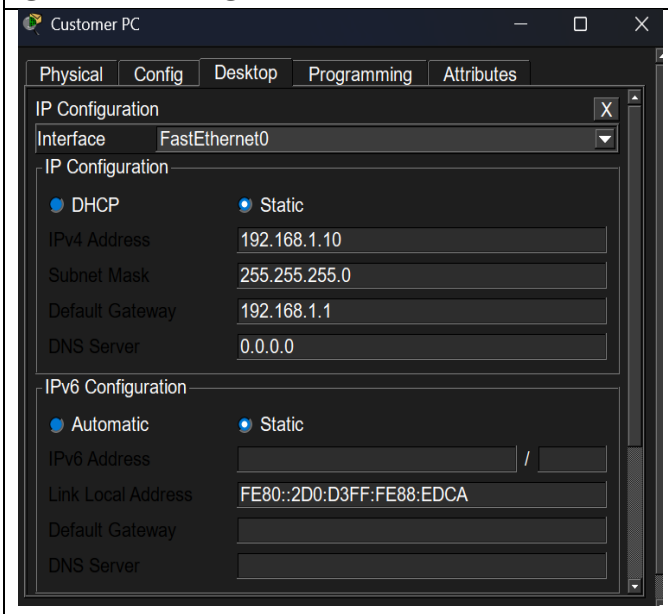
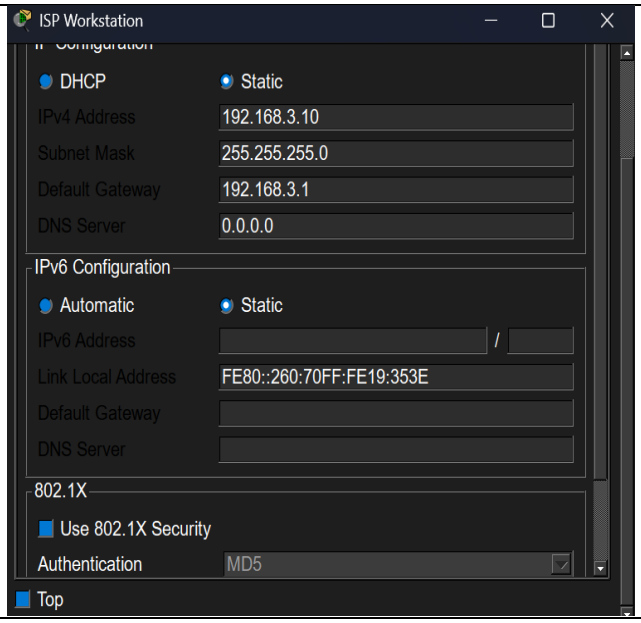
%LINK-5-CHANGED: Interface Serial0/1/0, changed state to down
IS-Router(config-if)#
IS-Router(config-if)#exit
IS-Router(config)#ip route 192.168.1.0 255.255.255.0 192.168.2.1
IS-Router(config)#end
IS-Router#
%SYS-5-CONFIG_I: Configured from console by console

IS-Router#write
Building configuration...
[OK]
```

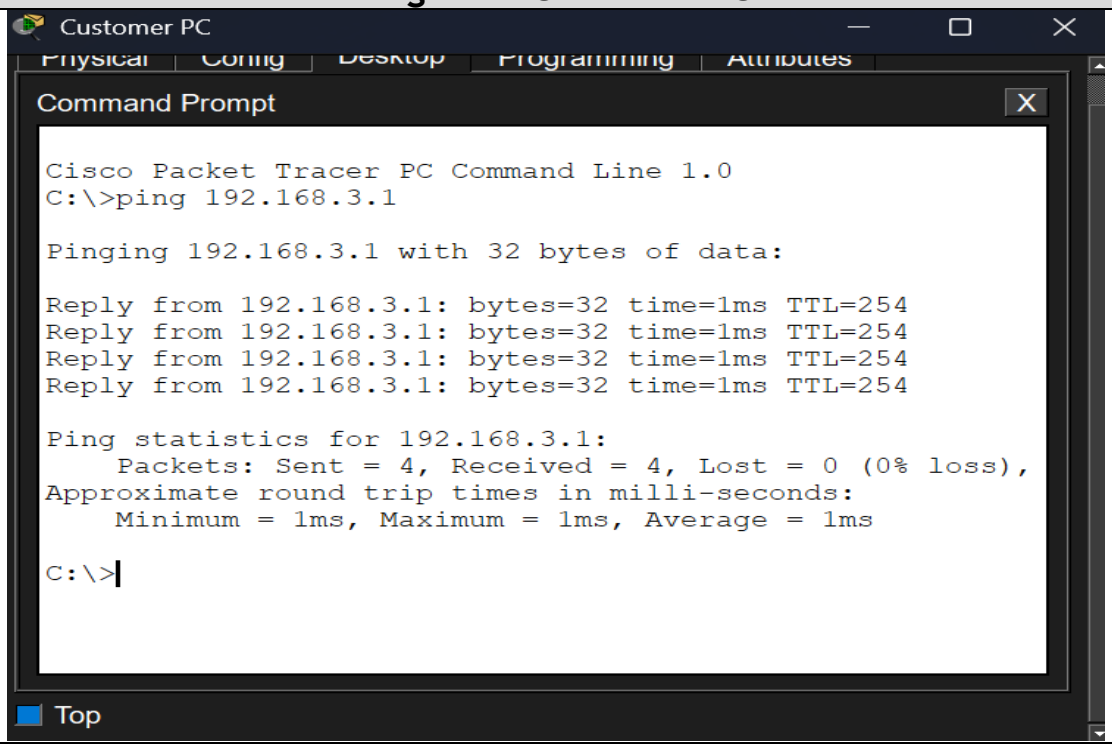
Verify the Customer-router configuration

```
Router-ISP#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        192.168.3.1     YES manual up                    up
FastEthernet0/1        unassigned      YES unset  administratively down down
Serial0/1/0            192.168.2.2     YES manual up                    up
Serial0/1/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router-ISP#show running-config
Building configuration...

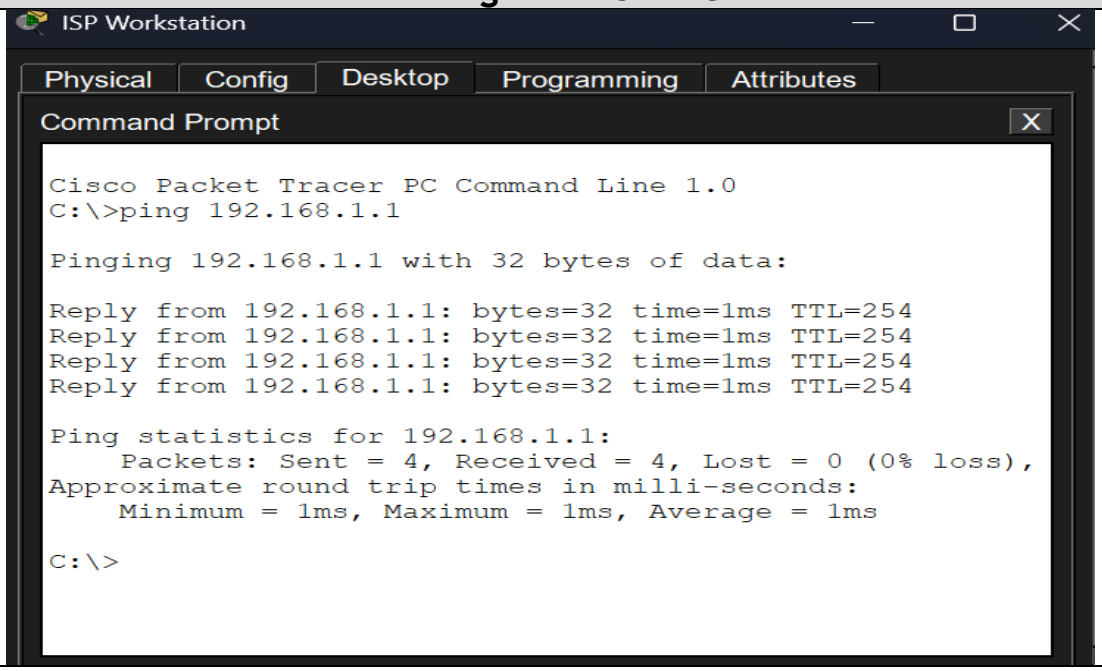
Current configuration : 937 bytes
!
version 12.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router-ISP
!
!
!
enable secret 5 $1$mERr$bZWscJx1wOHsv/xTRTJ8P.
enable password 1234
!
!
!
!
!
!
no ip cef
no ipv6 cef
interface FastEthernet0/0
 ip address 192.168.3.1 255.255.255.0
 duplex auto
 speed auto
!
interface FastEthernet0/1
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Serial0/1/0
 ip address 192.168.2.2 255.255.255.0
 clock rate 64000
!
interface Serial0/1/1
 no ip address
 clock rate 2000000
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
ip classless
ip route 192.168.1.0 255.255.255.0 192.168.2.1
!
ip flow-export version 9
!
!
!
banner motd ^C
!!Authorized access only!! ^C
line con 0
 password 1234
!
line aux 0
!
line vty 0 4
 password 1234
 login
!
!
!
end
```

Customer PC	ISP PC
	

5. Ping Customer PC to Workstation PC is it pinging

Ping from Customer PC


Ping from ISP PC



Conclusion:

Basic security configuration including hostnames, passwords, and custom banner messages was successfully applied to both network routers. Technical interface issues were resolved by assigning the IP configurations to the active Serial 0/1/0 interfaces instead of the default uninstalled slots. The serial link connection error was fixed by ensuring the clock rate command was placed precisely on the designated DCE side of the wire. Once all topology links turned green, static routing paths were correctly programmed on both devices to allow seamless data transfer across the subnets. Verification via an end-to-end ping from the Customer PC to the ISP Workstation confirmed excellent connectivity. The initial 25% packet drop was determined to be a standard, temporary ARP delay during physical address mapping. The entire network topology is now fully verified, secure, and operational for communication across all nodes.